D20A **Roll No. 58**

Experiment No. 3

Aim:

To create a **Cryptocurrency using Python** and perform **mining in the Blockchain created** using a simulated **three-node P2P blockchain network**.

Theory:

1. Blockchain

A **blockchain** is a distributed and decentralized ledger that stores information in a series of linked blocks. Each block contains transaction data, a timestamp, the previous block's hash, and its own hash. Since blocks are linked using hashes, modifying a previous block affects the subsequent blocks.

2. Mining

Mining is the process of collecting pending transactions into a block and solving a computational puzzle using **Proof-of-Work (PoW)**. Once a valid proof is found, the block is added to the blockchain and broadcast to connected peers. The miner receives a cryptocurrency reward for successfully mining the block.

3. P2P Blockchain Network

In this experiment, three blockchain nodes are used:
Node 1 → Port 5001
Node 2 → Port 5002
Node 3 → Port 5003
Each node maintains its own copy of the blockchain and communicates with other nodes to share and validate blocks.

4. Transactions

A transaction represents the transfer of cryptocurrency from one user to another. Each transaction contains:

- **Sender**
- **Receiver**
- **Amount**

Pending transactions are collected and added to a block during mining. A mining reward transaction is also added for the miner.

5. Proof-of-Work

Proof-of-Work requires the miner to perform computational calculations to find a valid proof. In the provided implementation, the hash generated using SHA-256 must satisfy the required condition before the block can be mined.

6. Longest Chain Rule

If different nodes have different versions of the blockchain, the longest valid chain is selected. The /replace_chain function obtains chains from connected peers and replaces the current

chain if a longer valid chain is found. This keeps the blockchain synchronized across the network.

7. Tools and Libraries

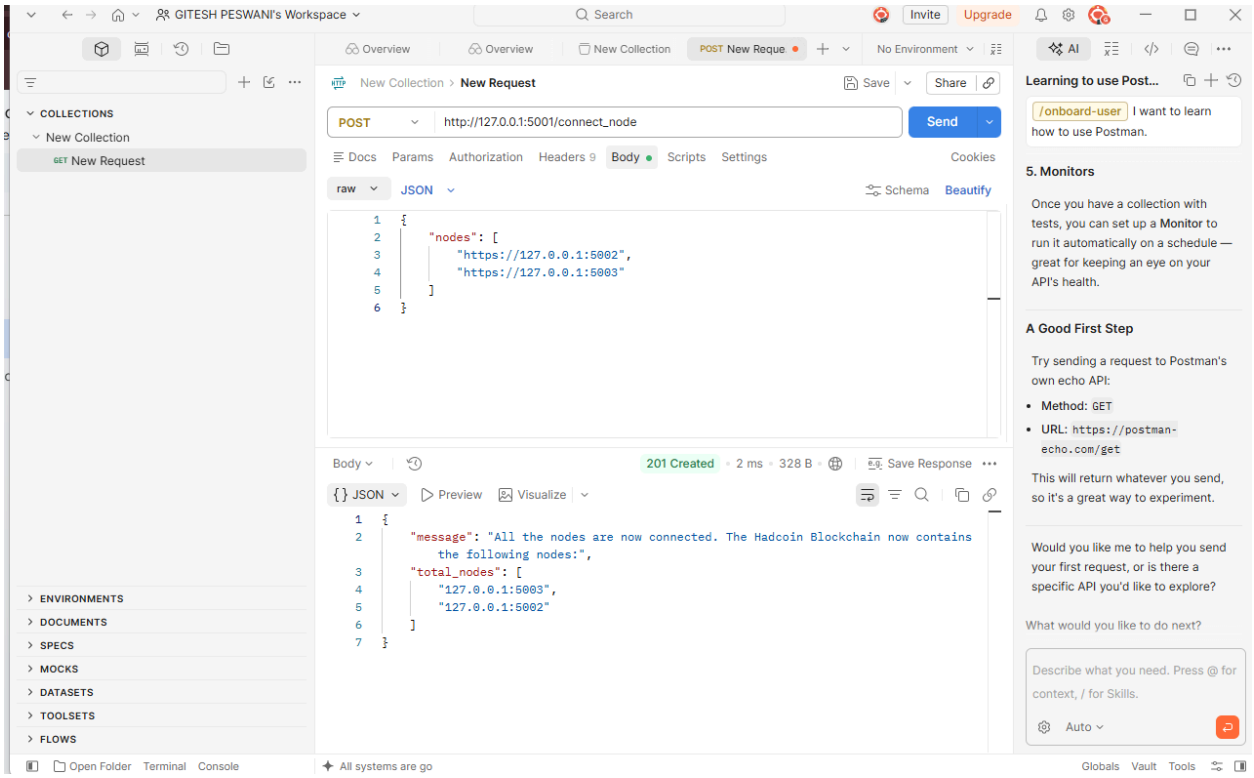
The experiment uses:

- **Python** - Programming language
- **Flask** - Creates blockchain API endpoints
- **Postman** - Sends and tests API requests
- **Requests** - Communication between blockchain nodes
- **Hashlib** - SHA-256 hashing
- **Datetime** - Timestamp generation
- **UUID** - Generates unique identifiers
- **URL parsing/request libraries** - Handling network requests

These tools and libraries are listed in the provided experiment material.

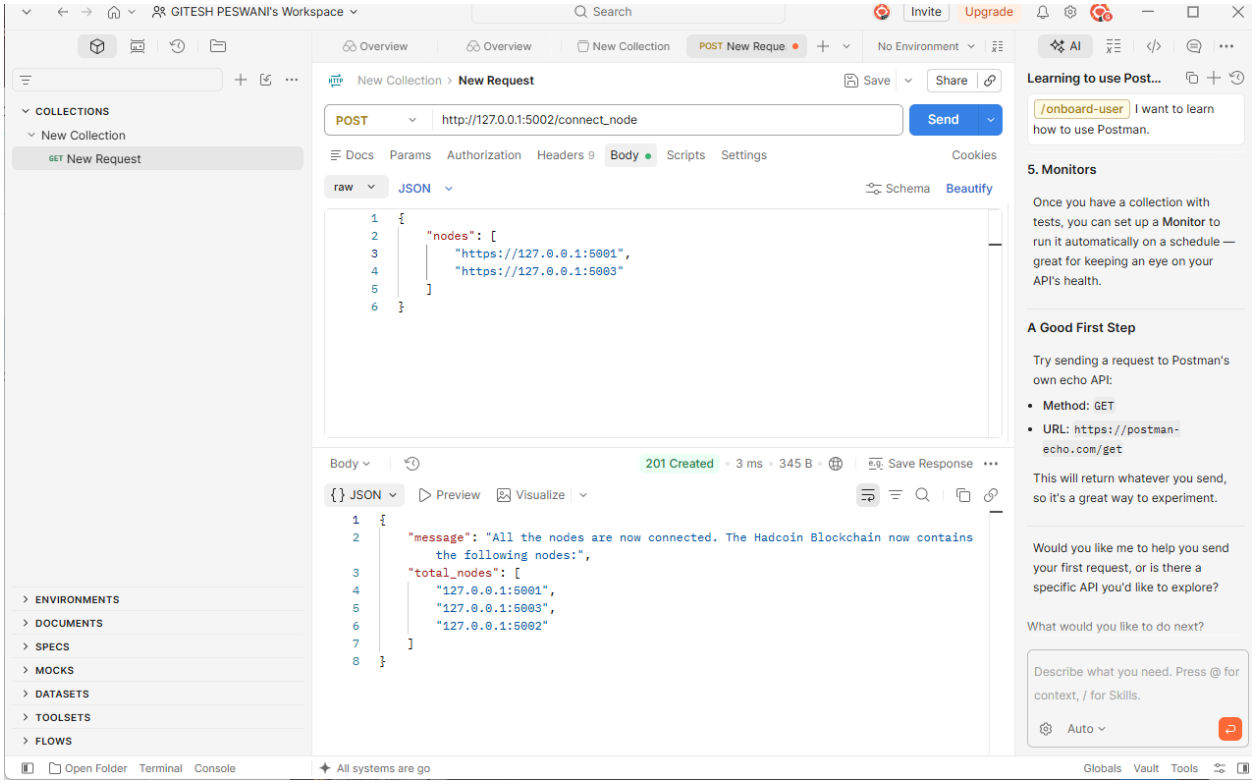
Output:

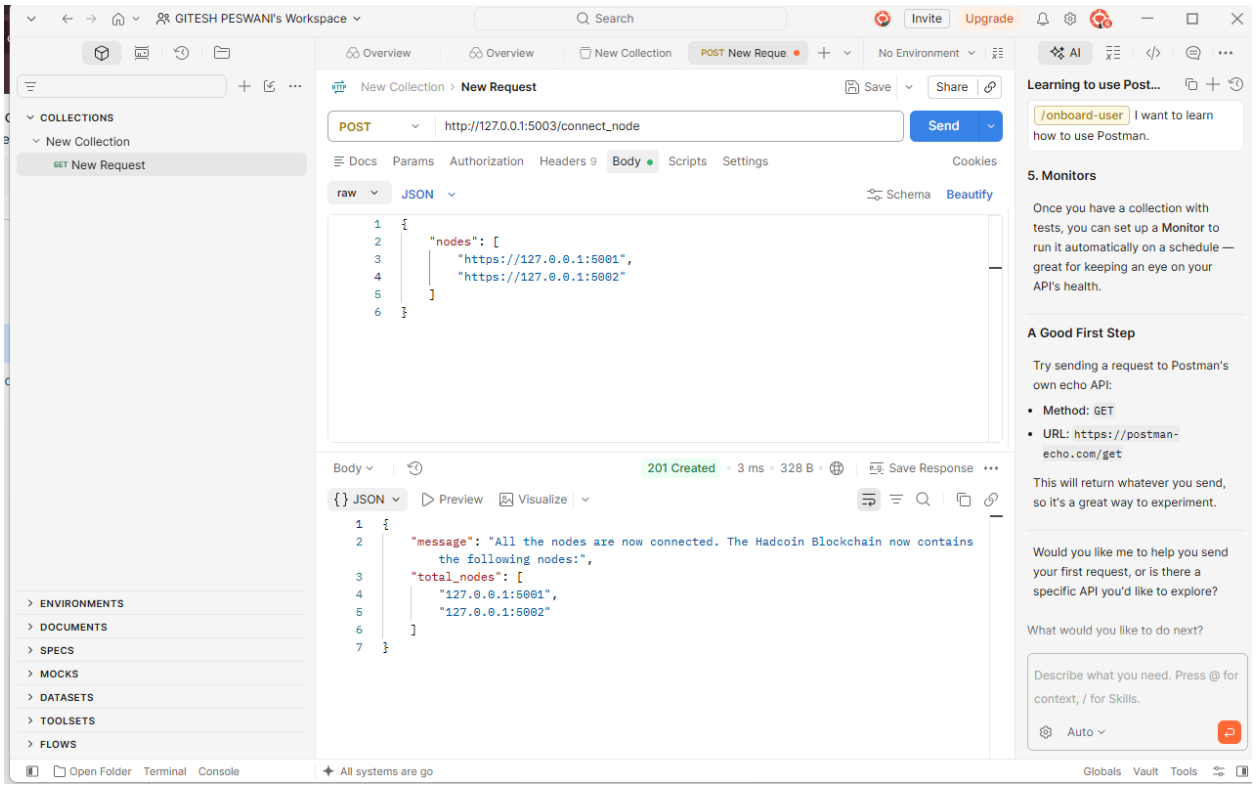
1. Nodes Connected :



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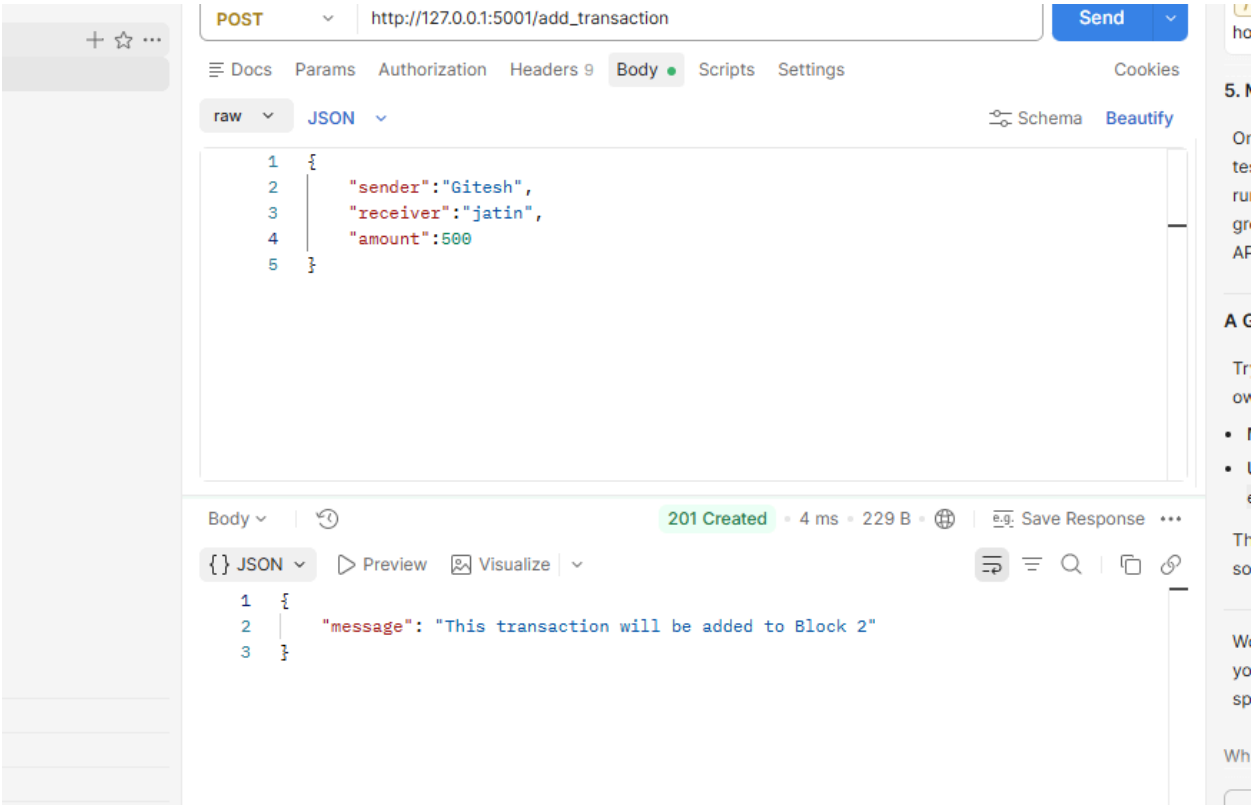
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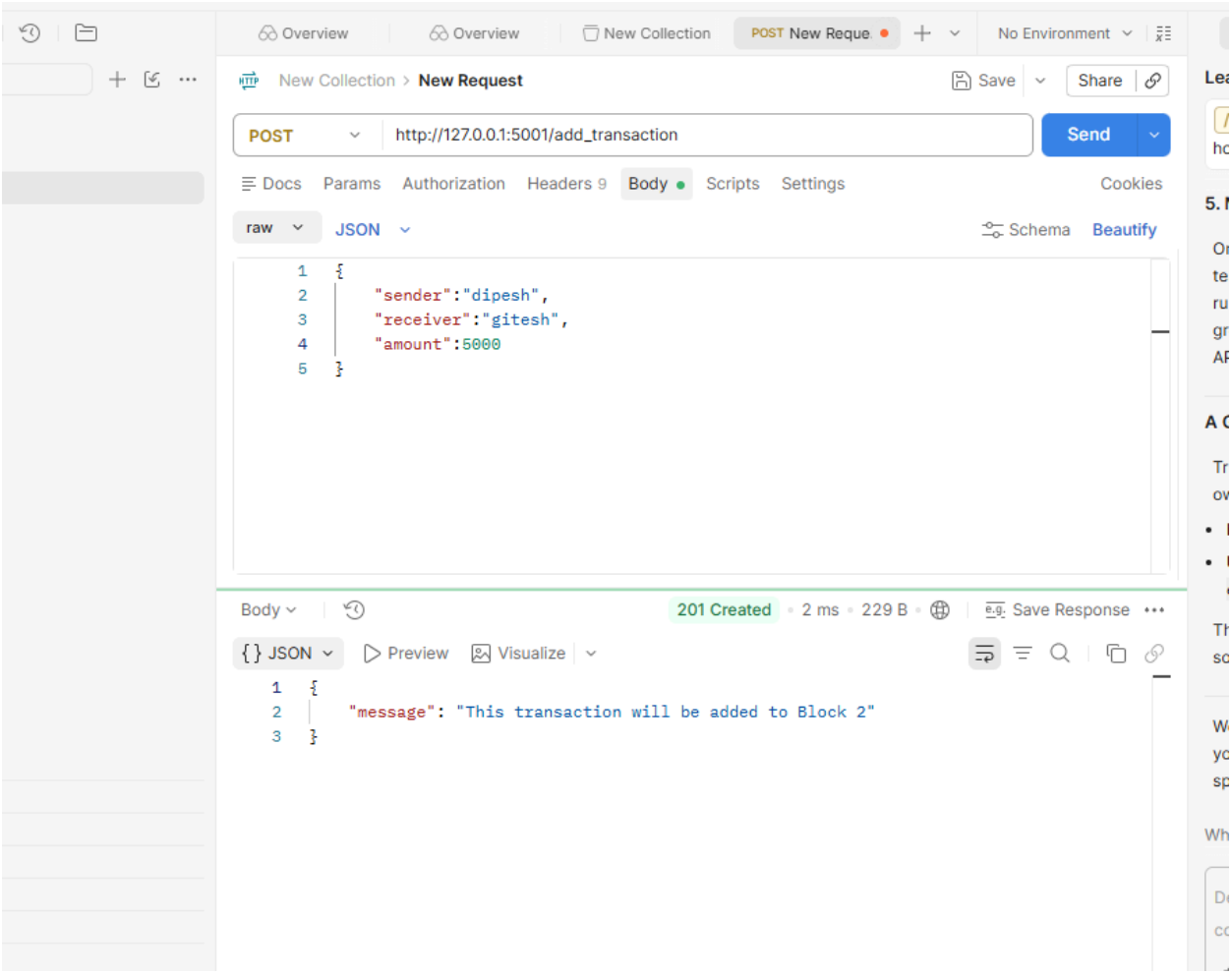
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2. Transactions Added :



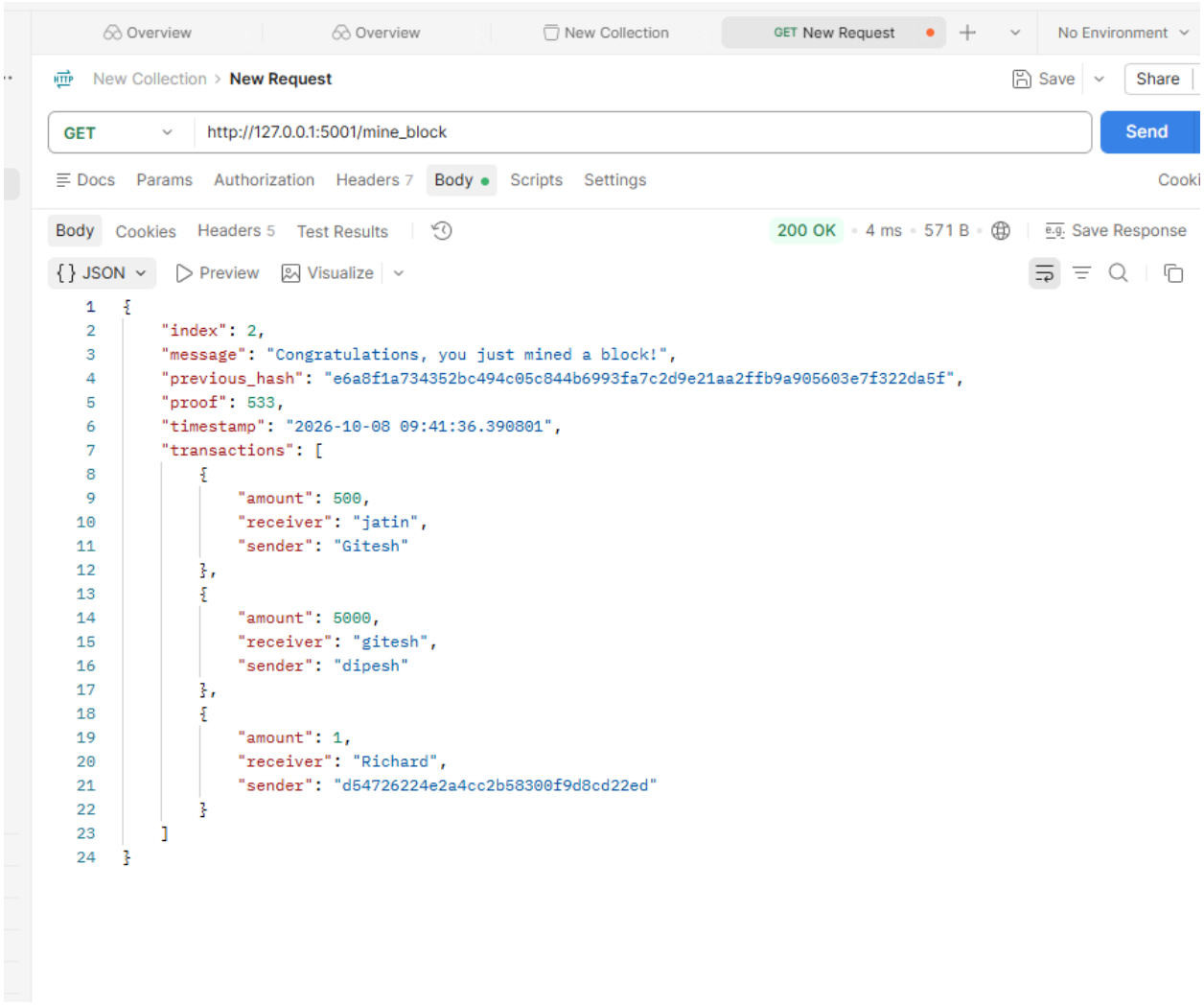
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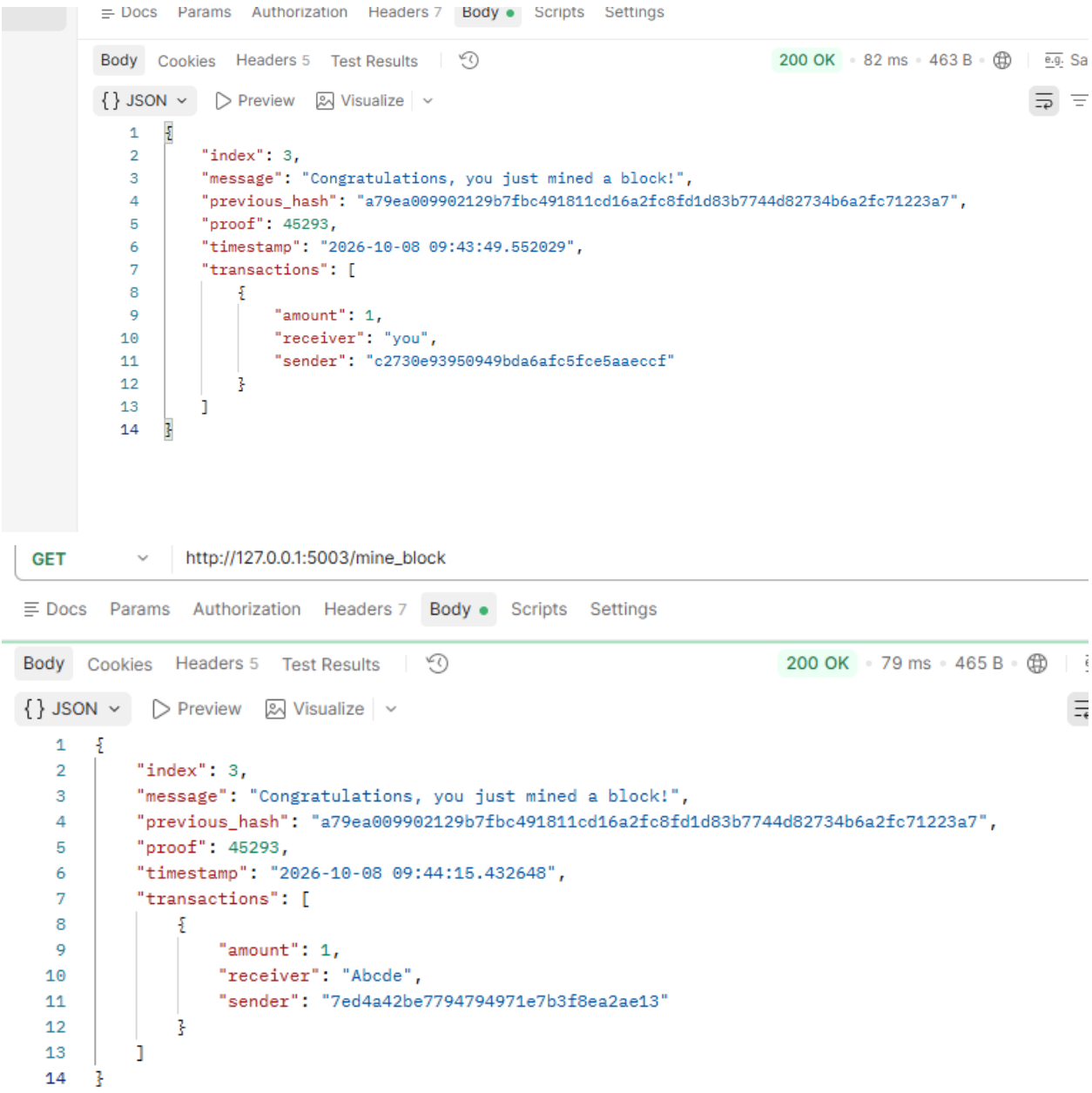


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3. Block Mined :



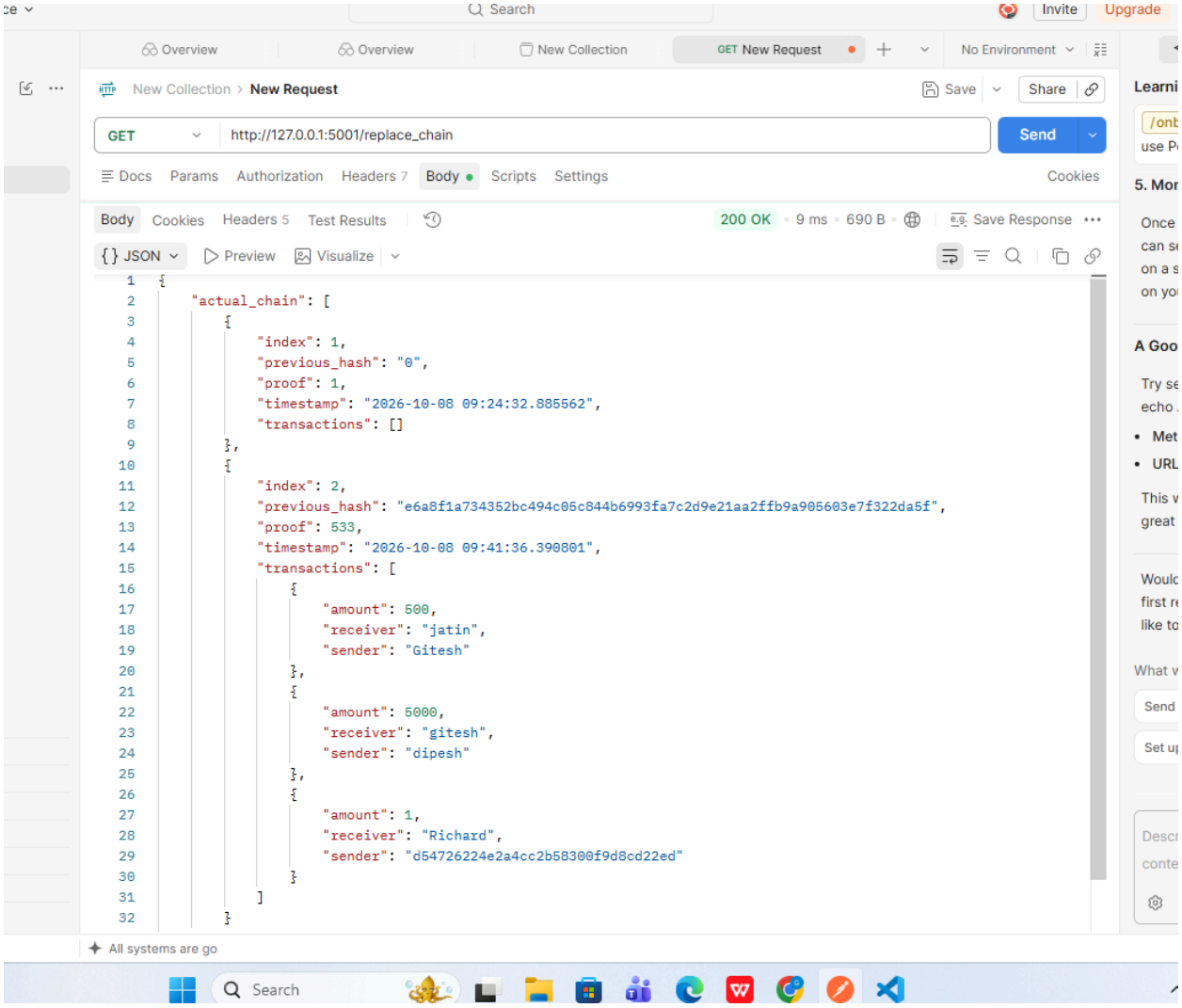
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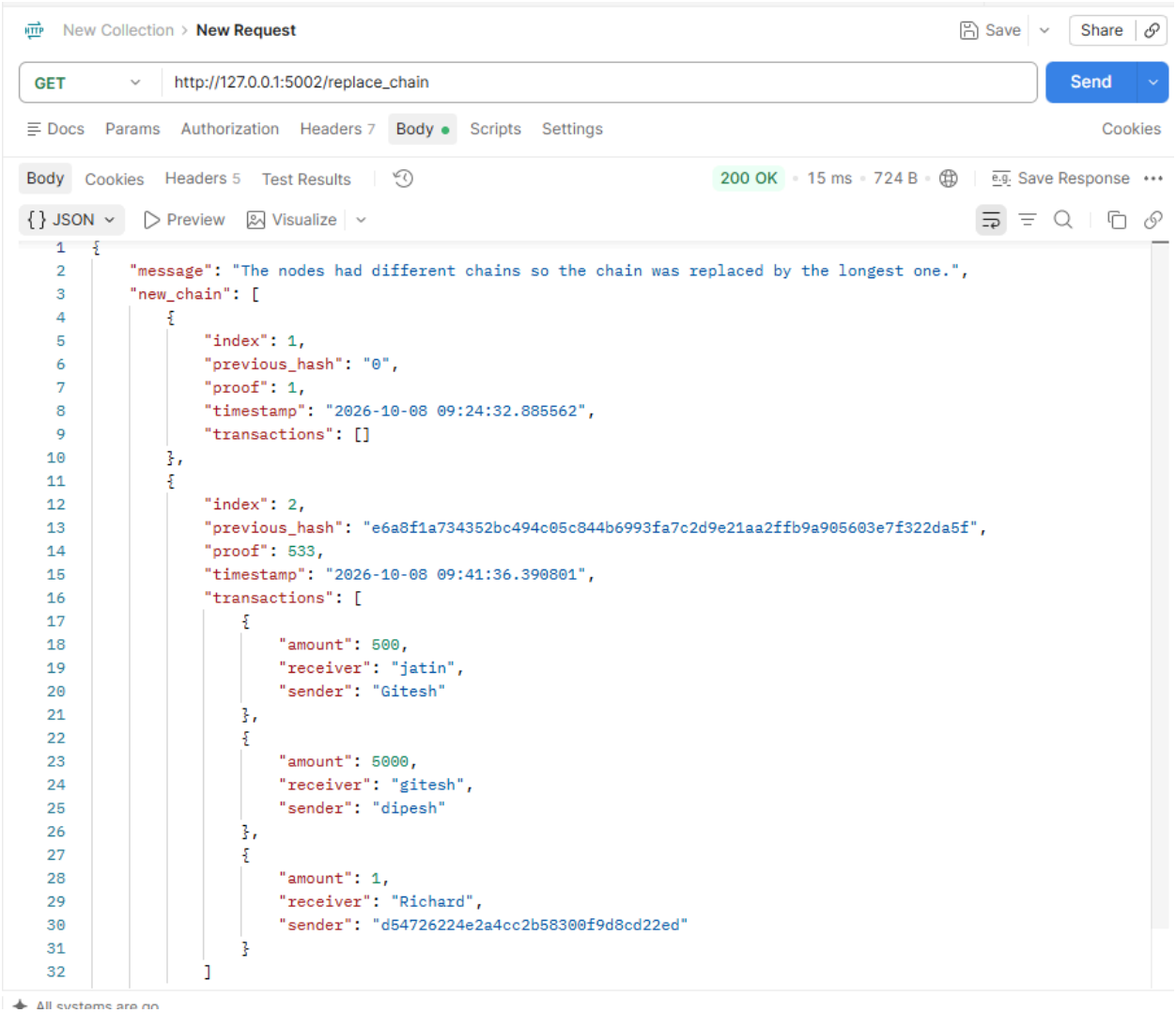
4. Chain Replacement :

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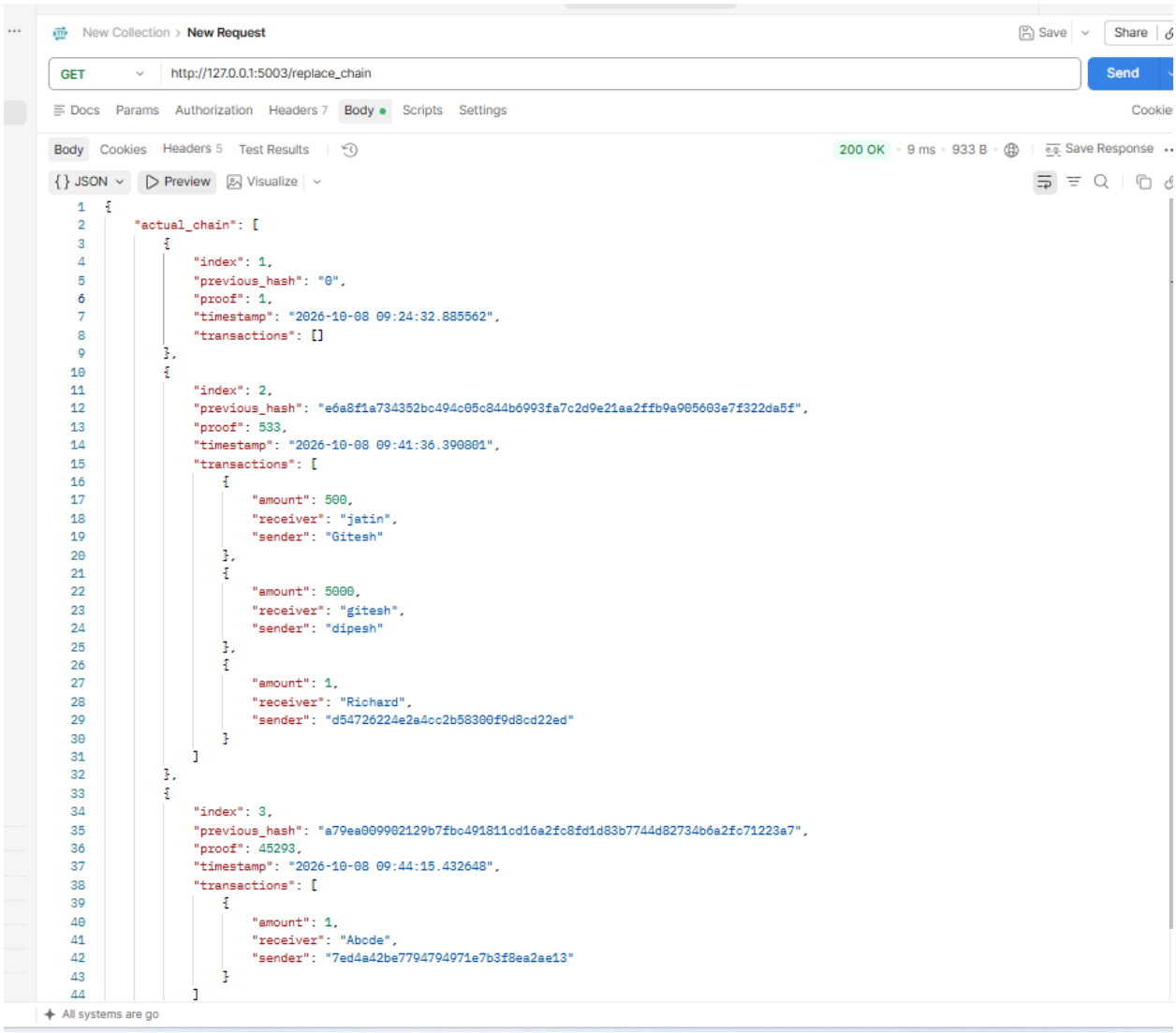
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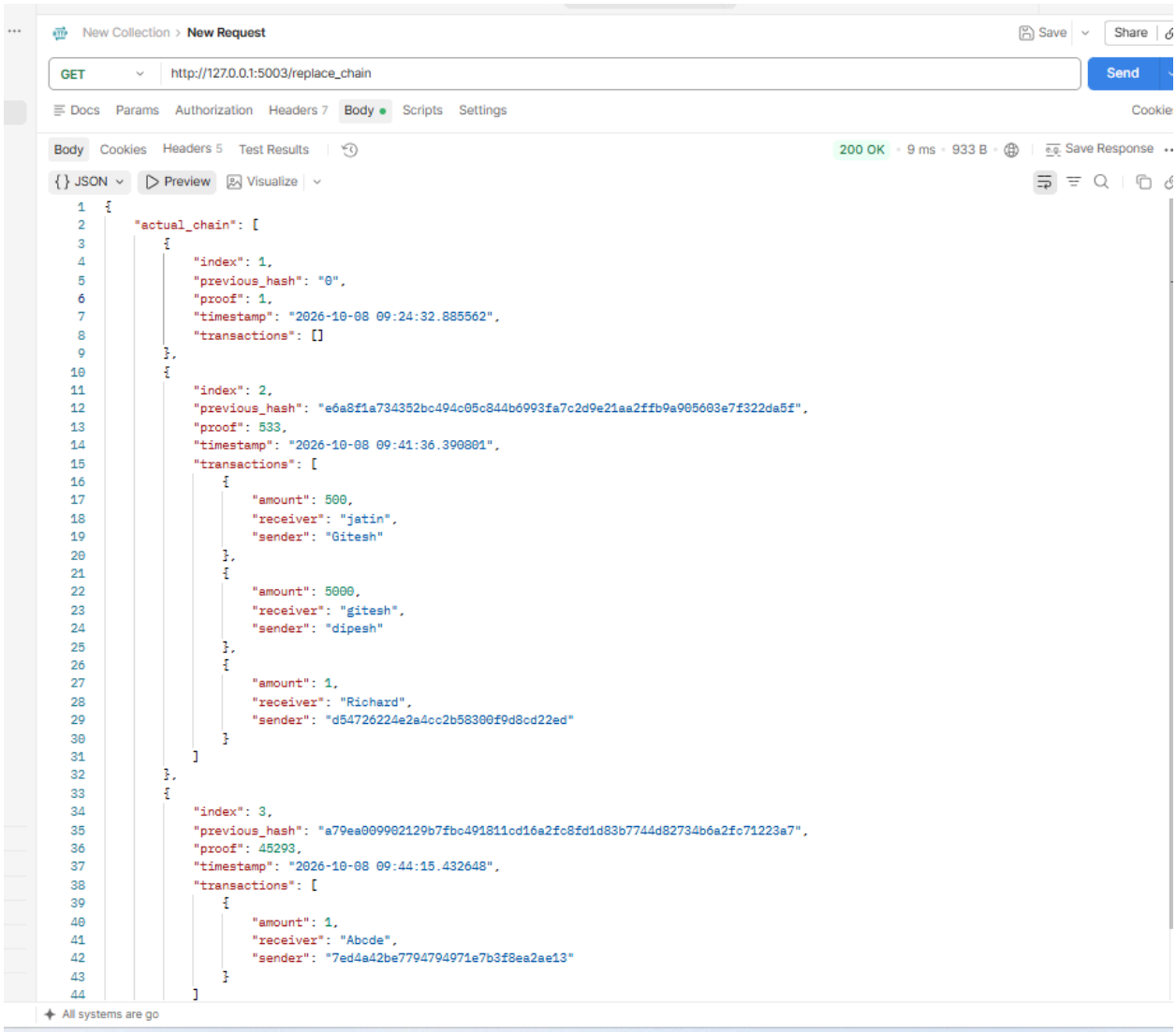
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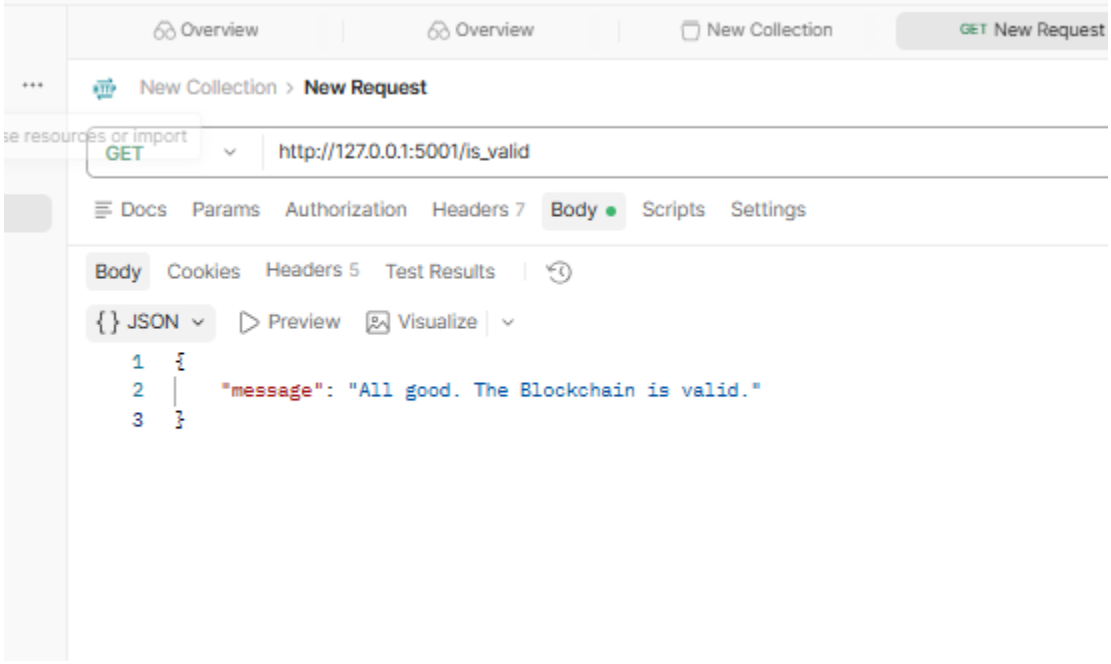


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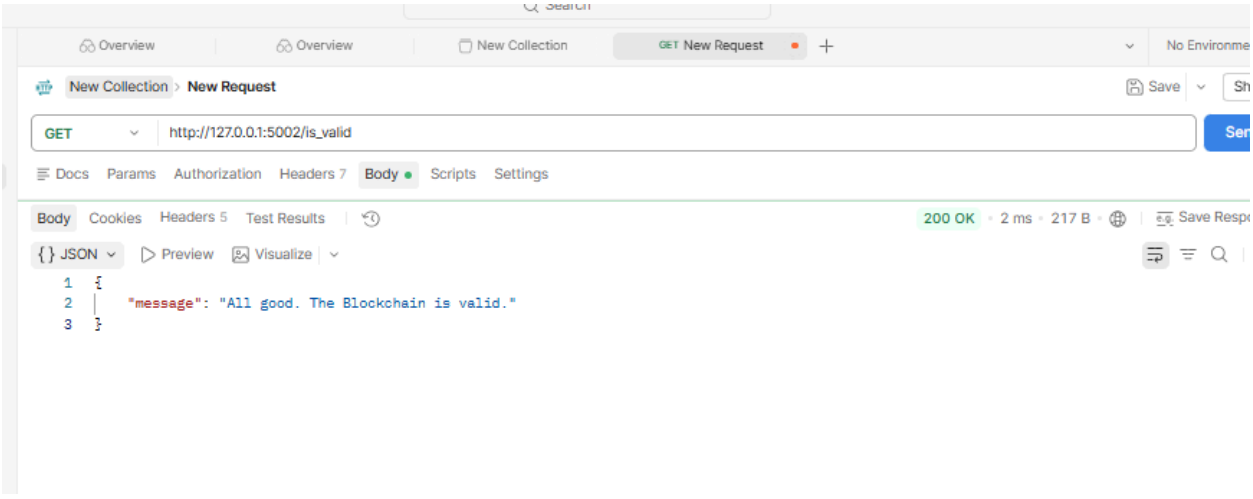
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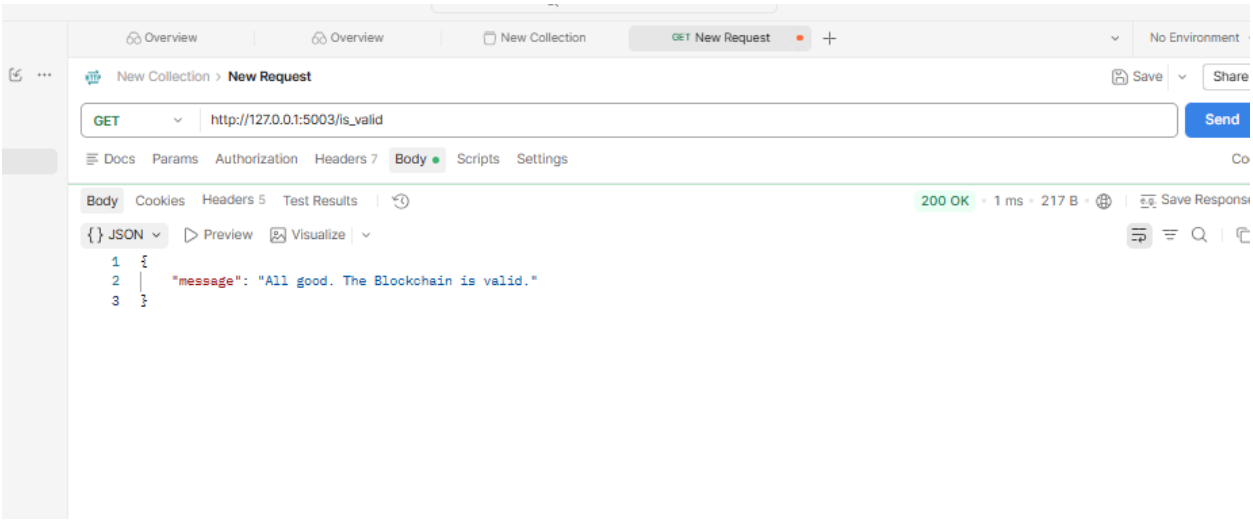
5. Blockchain Validation :



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CONCLUSION:

A cryptocurrency was successfully implemented using Python, Flask and a three-node P2P blockchain network. Transactions were added and mined using Proof-of-Work, and the blockchain was synchronized between nodes using the Longest Chain Rule. This experiment demonstrated cryptocurrency transactions, mining, decentralized consensus and blockchain network synchronization.